

What Speech Reveals: Interpreting Vocal Features and Evaluating Model Complexity for ASD Severity Prediction*

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Abstract— Speech-based analysis offers a scalable, non-invasive approach for ASD assessment, yet which vocal features matter most and whether model complexity is necessary remain incompletely understood. This study identifies the most relevant vocal features for ASD severity prediction, interprets their physiological and behavioral significance, and evaluates whether increased model complexity translates into performance gains, with a focus on lightweight solutions. Using the ASDSpeech dataset (258 ADOS-2 sessions from 197 children), 49 features spanning pitch, formants, energy, and others were extracted. Spearman correlation analysis reveals that pitch and spectral slope features exhibit the strongest associations with severity, particularly for Social Affect (SA), interpreted as reflecting a breathy phonation pattern linked to incomplete glottal closure. Formant features suggest atypical but highly repetitive vocal tract configurations, connecting to the restricted and repetitive behavior dimension of ASD. For prediction, a convolutional neural network (CNN) was compared with XGBoost across regression and classification tasks. XGBoost achieves comparable or slightly superior performance while requiring only 1–2% of the CNN's training and inference time, and proves more robust to feature removal. These findings suggest that feature selection and interpretation matter more than model complexity in speech-based ASD severity prediction.

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition defined by persistent deficits in social communication and interaction, alongside restricted, repetitive patterns of behavior [1]. Given its rising prevalence and systemic impact, ASD has become a focal point of clinical inquiry. Current estimates suggest ASD affects approximately 1% of the global population, a trajectory that necessitates the development of more efficient intervention frameworks [2].

Despite intensive research, diagnosis remains anchored in clinician-administered behavioral assessments. While these methods constitute the gold standard, they are inherently labor-intensive and susceptible to subjective bias [3], [4]. This limitation has catalyzed interest in objective, scalable, and technology-assisted modalities that augment clinical judgment by providing data-driven insights into behavioral patterns warranting further evaluation.

Speech represents a particularly salient modality in this context. Individuals with ASD frequently manifest distinct phonetic and prosodic signatures, the former encompassing the acoustic and articulatory properties of discrete speech sounds, while the latter captures suprasegmental elements such as pitch and rhythm. These features are linked to underlying

pathophysiological processes and serve as potential digital biomarkers [4]–[9].

Specifically, speech-based analytical tools may offer distinct advantages for clinical workflows, such as:

- Clinical augmentation: Refinement of subjective assessments by highlighting latent patterns, thereby improving consistency and alleviating clinician burden.
- Longitudinal monitoring: Facilitation of continuous, more-natural data collection across diverse settings.
- Granular progress tracking: Provision of objective metrics to monitor behavioral shifts over time.

For longitudinal tracking, lightweight and wearable-compatible systems are optimal. However, the existing literature fails to identify a consistently superior computational framework, with most models dominated by deep learning, which is notorious for its heavy computational footprint. Additionally, performance remains contingent upon feature representation, which also affects the computational demand of a model [6].

This study contributes toward understanding these discrepancies through a two-fold investigation: first, by evaluating commonly used speech features and their individual contributions to model performance, and second, by quantifying the level of model complexity required for ASD severity categorization. In doing so, it aims to support the development of rapid and reliable ASD detection tools suitable for deployment on handheld devices, with fast response times.

II. BACKGROUND

The Autism Diagnostic Observation Schedule, Second Edition (ADOS-2), is the gold-standard assessment for ASD. It is a semi-structured, clinician-administered evaluation that elicits social, communicative, and behavioral responses through standardized activities, which are scored based on observed behaviors. ADOS-2 generates scores for:

- Social Affect (SA) (0–22): Quantifies impairments in social communication and reciprocal interaction.
- Restricted and Repetitive Behaviors (RRB) (0–8): Captures the presence and intensity of stereotyped behaviors and interests.

Their sum yields the total ADOS score (0–30), which provides an overall estimate of severity, with higher scores indicating greater impairment.

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Utilizing speech as a biomarker for ASD is inherently complex, given the intricate physiological processes involved in speech signal generation and transmission. Speech production is traditionally conceptualized through the source-filter model, where airflow from the lungs facilitates vocal fold vibration at the larynx. This laryngeal activity constitutes the source, primarily dictating the fundamental frequency (F_0) and the underlying harmonic structure. The supraglottic vocal tract, comprising the pharyngeal, oral, and nasal cavities, functions as the filter, which reshapes the source signal by establishing a specific formant structure through resonance.

These source and filter components are fundamentally interdependent. The characteristics of the source, such as F_0 and phonation type, define the spectral envelope that enters the vocal tract, thereby modulating how specific formants are articulated. Conversely, the geometric configuration of the vocal tract exerts acoustic loading on the vocal folds; this back-pressure can reconfigure vibration patterns, subsequently modifying F_0 and the resulting voice quality. This bidirectional coupling underscores the difficulty of isolating discrete speech features, as their interdependence makes it challenging to attribute observed variability in ASD speech to a specific stage of the speech production process, with deviations potentially arising from both laryngeal and articulatory sources [10], [11].

Extensive research has identified a spectrum of speech features associated with ASD, with the most robust findings centered on prosody, specifically pitch mean and variance [6], [8]. A secondary tier of features includes temporal and intensity-based metrics, such as speech rate, duration, and loudness [6]. Although prevalent in the literature, these measures demonstrate inconsistent discriminative power and lack uniform application across studies. Beyond these core categories, feature selection remains highly fragmented; diverse investigations incorporate voice quality metrics (e.g., jitter, shimmer), spectral coefficients, and higher-level conversational dynamics such as vocalization frequency [6], [8].

This lack of standardization extends to the computational architecture; the field utilizes an array of modeling techniques ranging from traditional machine learning—such as Support Vector Machines (SVM), k-means clustering, and Random Forests—to complex deep learning frameworks, including Convolutional and Dense Neural Networks (CNN/DNN[5], [8], [12]).

It remains a research challenge to determine which approaches and features best balance diagnostic sensitivity with the computational efficiency needed for long-term tracking.

III. METHODS

A. Dataset

The data used in this study were obtained from the ASDSpeech dataset [5], collected at a national autism research center and multiple affiliated clinical diagnostic sites. Recordings were acquired during standardized ADOS-2 assessments conducted in controlled clinical environments. All participants received confirmed ASD diagnoses.

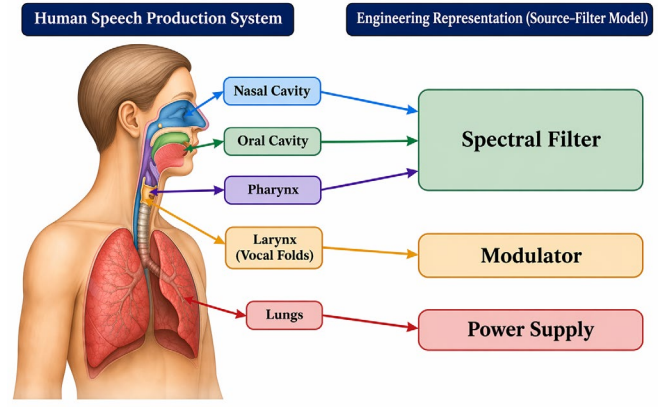


Figure 1. Source-filter representation of speech production

The cohort includes 197 children (aged 1–7 years) recorded between 2015 and 2021, yielding a total of 258 assessment sessions. While 136 children participated in a single assessment session, 61 underwent longitudinal assessments across two separate sessions, separated by 10–29 months.

B. Data acquisition and feature extraction

Audio was recorded in clinical assessment rooms using a wall-mounted microphone positioned approximately 1–2 meters from the child. Signals were captured at a sampling rate of 44.1 kHz with 16-bit resolution and subsequently downsampled to 16 kHz for analysis. Each session lasted approximately 40 minutes and included both structured tasks and spontaneous vocalizations, reflecting realistic clinical interaction conditions.

Child vocalizations were manually annotated to isolate speech segments corresponding exclusively to the child, excluding clinician speech. The annotated segments were further subdivided into individual vocalizations using energy-based thresholding relative to the estimated background energy E_b . A vocalization onset was detected when the short-time signal energy exceeded $E_b + 2.7 \text{ dB}$ for a minimum duration of 50ms, while the offset was defined when the energy fell below $E_b + 0.4 \text{ dB}$ for at least 50ms. To ensure reliable feature extraction, each detected vocalization was required to have a minimum duration of 110ms.

Feature extraction was performed at the vocalization level using short-time analysis. Low-level descriptors were first computed for each vocalization. These descriptors were then converted into higher-level statistics across sequences of 10 consecutive vocalizations. This approach produced vocalization-to-vocalization feature representations that capture fluctuations and dynamics in speech behavior over short temporal contexts. The extracted feature set is summarized in Table I.

These features primarily capture prosodic properties of speech that can be analyzed with reduced sensitivity to their underlying linguistic content, though some, particularly formants, remain closely tied to phonemic structure. Pitch (F_0) refers to the perceived highness or lowness of the voice and reflects how fast the vocal folds vibrate. Features based on F_0 describe both its average value and how much it varies

TABLE I. SPEECH FEATURES USED IN THIS WORK

#	Feature type	Feature
1	Pitch (F0)	Mean F0 across consecutive vocalizations
2		Variance of F0 across consecutive vocalizations
3		Correlation of coefficient (σ^2/μ) across consecutive vocalizations
4		Mean of minimum F0 across consecutive vocalizations
5		Variance of minimum F0 across consecutive vocalizations
6		Mean of maximum F0 across consecutive vocalizations
7		Variance of maximum F0 across consecutive vocalizations
8		Mean F0 across VVs within consecutive vocalizations
9		Variance of F0 across VVs within consecutive vocalizations
10		Variance of mean F0 across VVs
11	Formants & BW	Mean F1 across consecutive vocalizations
12		Variance of F1 across consecutive vocalizations
13		Mean F2 across consecutive vocalizations
14		Variance of F2 across consecutive vocalizations
15		Mean F1-F2 across consecutive vocalizations
16		Variance of F1-F2 across consecutive vocalizations
17		Mean BW1 across consecutive vocalizations
18		Variance of BW1 across consecutive vocalizations
19		Mean BW2 across consecutive vocalizations
20		Variance of BW2 across consecutive vocalizations
21	Jitter	Mean jitter across consecutive vocalizations
22		Variance of jitter across consecutive vocalizations
23	Voicing	Mean voicing across consecutive vocalizations
24		Variance of voicing across consecutive vocalizations
25	Energy	Mean E across consecutive vocalizations
26		Mean ΔE across consecutive vocalizations
27		Mean $\Delta\Delta E$ across consecutive vocalizations
28		Mean ΔE across consecutive vocalizations
29		Variance of E across consecutive vocalizations
30		Variance of ΔE across consecutive vocalizations
31		Variance of $\Delta\Delta E$ across consecutive vocalizations
32		Variance of ΔE across consecutive vocalizations
33	ZCR	Mean ZCR across consecutive vocalizations
34		Variance of ZCR across consecutive vocalizations
35		Mean ZCR across VVs
36		Variance of ZCR across VVs
37		Mean ZCR across UVs
38		Variance of ZCR across UVs
39	Spectral slope (SpSl)	Mean SpSl (20–500 Hz)
40		Variance SpSl (20–500 Hz)
41		Mean SpSl (50–1500 Hz)
42		Variance SpSl (50–1500 Hz)
43		Mean SpSl (20–500 Hz) across VVs
44		Mean SpSl (20–500 Hz) across UVs
45		Mean SpSl (50–1500 Hz) across VVs
46		Mean SpSl (50–1500 Hz) across UVs
47	Duration	Mean duration across consecutive vocalizations
48		Variance of duration across consecutive vocalizations
49	Quantity	Total number of vocalizations

VV: Voiced Vocalizations, UV: Unvoiced Vocalizations

across consecutive vocalizations, capturing changes in intonation and expressiveness. Formants (F1 and F2) are resonant frequencies shaped by the vocal tract and are closely related to how sounds are articulated. The difference between F1 and F2, along with their bandwidths (BW), provides insight into vocal tract configuration. Jitter measures small, cycle-to-cycle variations in vocal fold vibration, while voicing reflects the presence of periodic vocal fold activity.

Energy (E) represents the intensity of the acoustic signal, a physical correlate of perceived loudness, while its derivatives (ΔE and $\Delta\Delta E$) capture how intensity changes over time, reflecting dynamics such as emphasis or abrupt transitions. The zero-crossing rate (ZCR) measures how frequently the signal changes sign and is often associated with the noisiness or sharpness of a sound. Spectral slope (SpSl) characterizes the rate at which energy decreases across frequencies, providing information about the balance between low- and high-frequency components in the voice. Duration features quantify how long vocalizations last and how consistent they are, while the total number of vocalizations reflects speech quantity. Full details are provided in the original work, along with its attached code [5].

To ensure robustness and coverage of recording variability, this process was repeated 1000 times per child using randomly sampled segments from the full recordings. Each sample consists of 10 sequential vocalizations used to generate the 49 features.

B. Feature analysis

A two-phase analysis quantified feature utility and structural relationships. Feature 49 (Quantity) was removed as irrelevant (minimal variation across children), and feature values were averaged across samples per session to yield a single representative vector per assessment.

First, Spearman rank correlations were computed between session-level features and ADOS-2 subscores (SA and RRB). This non-parametric measure was chosen for its ability to capture monotonic relationships without assuming linearity or normality. Second, the same metric was applied between features to assess their interdependency.

C. Classification

The sessions were partitioned into training, validation, and testing sets as follows. Single-session recordings formed the training set (136 sessions); for children with repeated assessments, the first and second sessions were assigned to validation and testing respectively (61 sessions each, same participants).

Twenty models were trained to assess robustness, each using 20 randomly selected samples per child, where each sample corresponds to one recording (10 sequential vocalizations). Two architectures were evaluated—CNN and XGBoost—across regression and classification tasks. For classification, continuous outcomes were discretized into two ordinal classes using the median as the threshold (to reduce sensitivity to extreme values). For regression, original continuous scores were used directly.

For the XGBoost model, the 20×48 feature map was first reduced to a one-dimensional representation by averaging

across the 20 samples per child. In contrast, the CNN operated directly on the full 20×48 input, where the convolutional layers learned how to aggregate information across samples. The architecture of the CNN is shown in Table II.

The CNN used a softmax output with sparse categorical cross-entropy for classification and a linear output with mean squared error for regression, optimized with RMSprop. All models were trained with a learning rate of 1×10^{-3} for up to 50 epochs using early stopping (patience = 20) based on validation performance, restoring the best weights. XGBoost was configured with 400 trees, max_depth = 4, learning_rate = 0.05, subsample = colsample_bytree = 0.8, with default regularization. Classification used the multi:softmax objective with two classes, while regression used reg:squarederror, with early stopping (early_stopping_rounds = 30) based on validation performance.

Experiments were conducted on a Windows 10 machine with Python 3.10.11, equipped with an Intel Core Ultra 9 285HX CPU, 32 GB RAM, and an NVIDIA RTX PRO 2000 Laptop GPU with 8 GB VRAM. The CNN was implemented with TensorFlow 2.10.0 and gradient boosting with XGBoost 3.2.0. Preprocessing and evaluation used scikit-learn 1.7.2.

TABLE II. ARCHITECTURE OF THE CNN MODEL USED FOR CLASSIFICATION AND REGRESSION TASKS.

Layer	Output Shape	Parameters	Activation
Input	(20, 48)	—	—
Conv1D (64, k=3)	(20, 64)	9,280	ReLU
MaxPooling1D (2)	(10, 64)	0	—
Conv1D (128, k=3)	(10, 128)	24,704	ReLU
Flatten	(1280)	0	—
Dense (128)	(128)	163,968	ReLU
Dropout (0.3)	(128)	0	—
Dense (64)	(64)	8,256	ReLU
Last Layer	Classification	(2)	130
	Regression	(1)	65
			Softmax
			Linear

Total parameters are 206,338 for classification and 206,273 for regression.

IV. RESULTS

A. Speech feature – severity score correlations

Figure 2 presents the Spearman rank correlation coefficient (ρ) and its statistical significance between speech features and severity assessment scores, stratified by sex (males, females, and combined) across the eight feature families described in Table I.

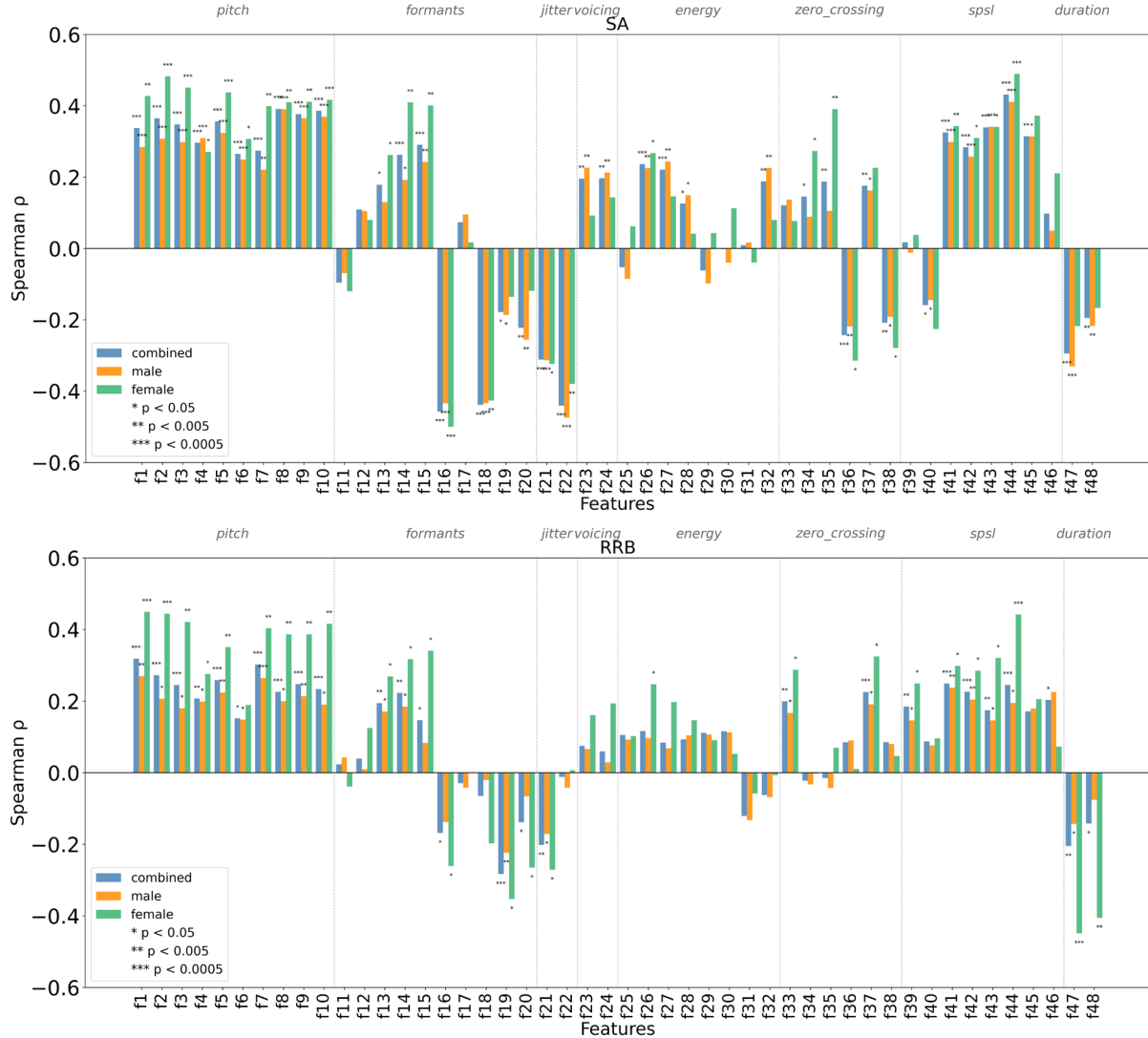


Figure 2. Spearman correlation coefficients between speech features and severity scores (SA and RRB)

Several patterns emerge across both SA and RRB. Strong positive correlations are primarily observed in pitch (F0) and spectral slope (SpSl), while strong negative correlations are evident in jitter and duration-related features. For formant-related features, positive correlations are observed for the mean and variance of F2 and the mean of $|F1-F2|$, whereas negative correlations are found for the variance of $|F1-F2|$ (SA only), and the mean and variance of BW2.

The combination of high F0 and steep spectral slope (measured between 50–1500 Hz and 20–500 Hz) suggests a vocal posture in which the vocal folds vibrate rapidly, producing elevated pitch, but fail to adduct fully during each vibratory cycle. This incomplete glottal closure allows excess airflow to escape, steepening the spectral tilt after the first harmonic. The result is a voice that sounds high-pitched yet thin and airy. For formant features, higher mean and variance of F2 suggest children with greater severity show more anterior vocal tract configurations on average and do so inconsistently, resulting in subtly fronted or atypical vowel realizations. Yet the lower variance of $|F1-F2|$ (SA only) and narrower, less variable BW2 reveal a paradox: despite atypical average vocal tract positioning, these configurations are produced in a highly repetitive, fixed manner — suggesting a vocal tract locked into a restricted set of shapes rather than ranging freely across the acoustic vowel space.

Overall, correlation magnitudes are more pronounced for SA than for RRB, likely due to SA's broader scale (0–22 versus 0–8) or the structured nature of the task eliciting behaviors more effectively captured by these features.

Correlations tend to be stronger in female participants, with combined results falling between the two sex-specific values. To assess whether this reflects sample imbalance (192 male vs. 66 female sessions), a bootstrap resampling procedure was applied to each subgroup, drawing 1,000 samples with replacement matched to the female sample size. Spearman's ρ and 95% confidence intervals were computed across iterations (Figure 3).

The results show that, for positively correlated features (e.g., pitch and spectral slope), the confidence intervals for females tend to lie slightly above those of males, while for negatively correlated features (e.g., jitter and duration), they tend to lie slightly below. This pattern suggests a generally stronger association in females compared to males. Although the differences are modest, they appear to be consistent and not merely attributable to dataset variability.

B. Feature-feature correlation

The second set of results examines the relationships among the different feature types. These inter-feature correlations are visualized using a heatmap, as shown in Figure 4.

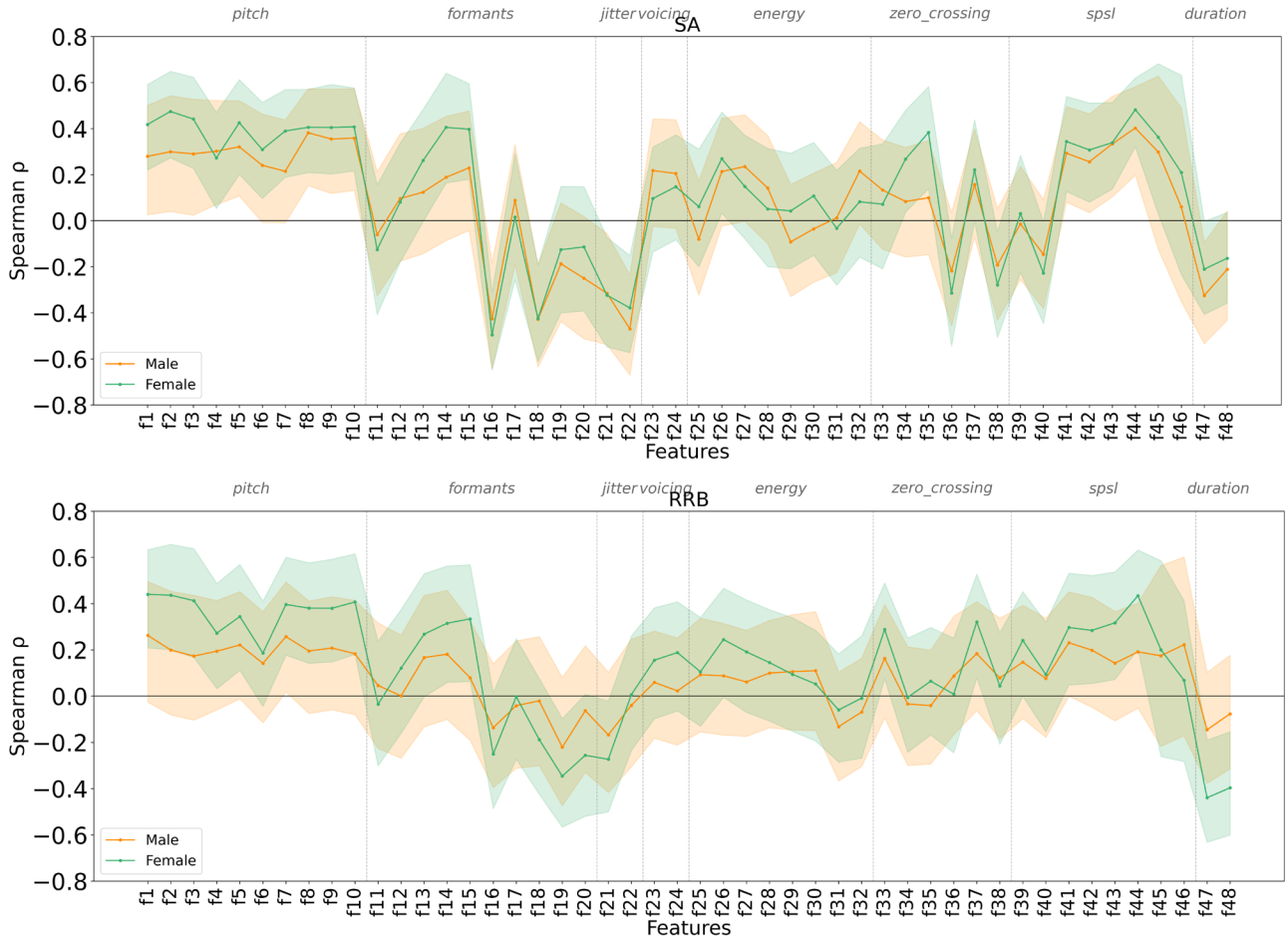


Figure 3. Bootstrap-estimated Spearman correlation (ρ) with 95% confidence intervals between speech features and severity scores (SA and RRB), stratified by sex.

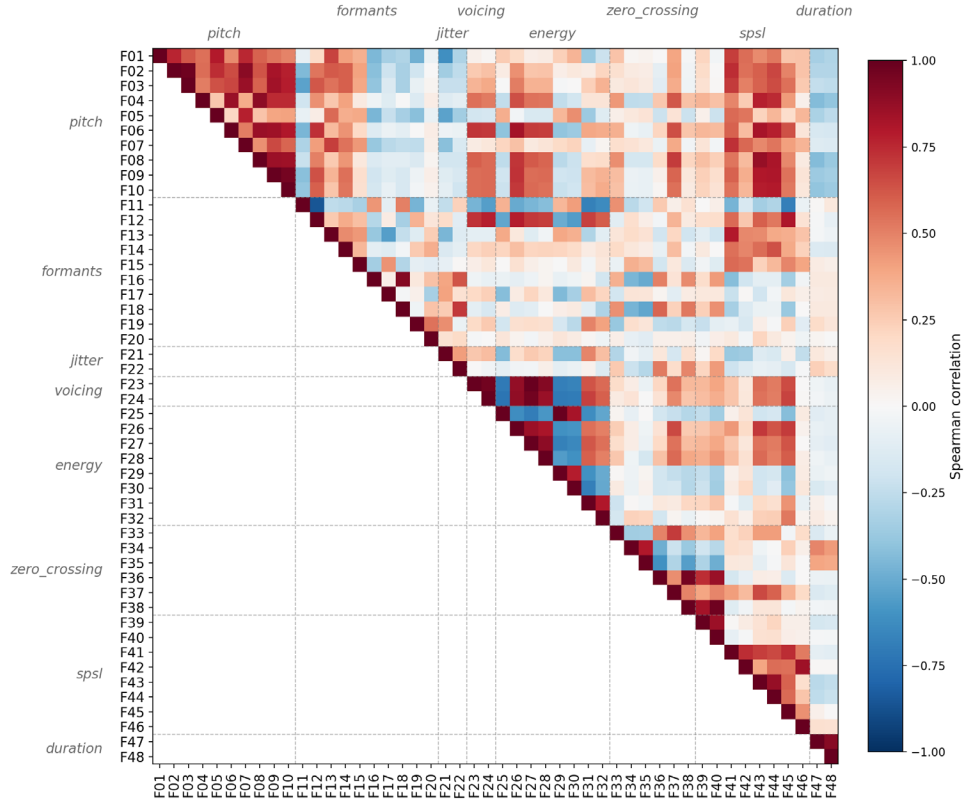


Figure 4. Spearman correlation heatmap of inter-feature relationships

The heatmap reveals strong intercorrelations within feature clusters, indicating considerable redundancy: pitch features are highly intercorrelated, as are some spectral slope features (F41–F46), energy features (F25–F32), and duration features (F47–F48), suggesting these groups largely capture overlapping information. Across clusters, pitch and spectral slope features show strong positive cross-correlations, reinforcing the breathy phonation interpretation: high F0 (F01, F04, F06) and steep spectral slope (F41–F46) are not independent signals but facets of the same underlying vocal behavior.

Duration features (F47–F48) show small negative correlations with pitch and spectral slope, suggesting that higher-pitched, breathier vocalizations tend to also be shorter, possibly reflecting reduced respiratory support or fragmented speech patterning. Jitter features (F21–F22) are negatively linked to pitch, consistent with higher F0 co-occurring with reduced cycle-to-cycle frequency perturbation.

Together, this pattern of within-cluster redundancy and between-cluster independence suggests that dimensionality reduction or careful feature selection would be advisable before modelling to avoid multicollinearity while preserving the breadth of information captured.

More broadly, formant-related features tend to exhibit weak correlations with features from other families, suggesting that they capture information not well represented by other feature groups.

C. Classification

Two models were evaluated: a CNN and XGBoost. XGBoost was included as a computationally cheaper alternative, but also because averaging input features across samples, as done here, likely reduces the CNN's primary advantage of learning local structure through convolution, since similar effects can be achieved through explicit feature averaging. The comparative results are presented in Figure 5. The annotations represent the difference from the baseline case where no feature families are dropped. Feature families were dropped one at a time to assess their individual impact on model performance.

XGBoost outperforms the CNN slightly in terms of NRMSE, with a maximum difference of around 0.04, and also proves more stable when feature families are removed — CNN performance is notably more sensitive to such changes. Interestingly, the CNN improves when formant and spectral slope features are excluded, despite both families showing strong correlations with severity scores, particularly for SA. This likely reflects overfitting rather than a genuine insensitivity to these features. XGBoost behaves more consistently with expectations: removing these families increases error, confirming their relevance. Based on the XGBoost results, pitch appears to be the most critical feature family for predicting SA, while formant-related features carry greater importance for RRB.

When classification was performed instead of regression to differentiate between severe and non-severe ASD scores, an accuracy of ~73% was achieved for the baseline case (no feature removal) in SA, and ~68% for the RRB case.

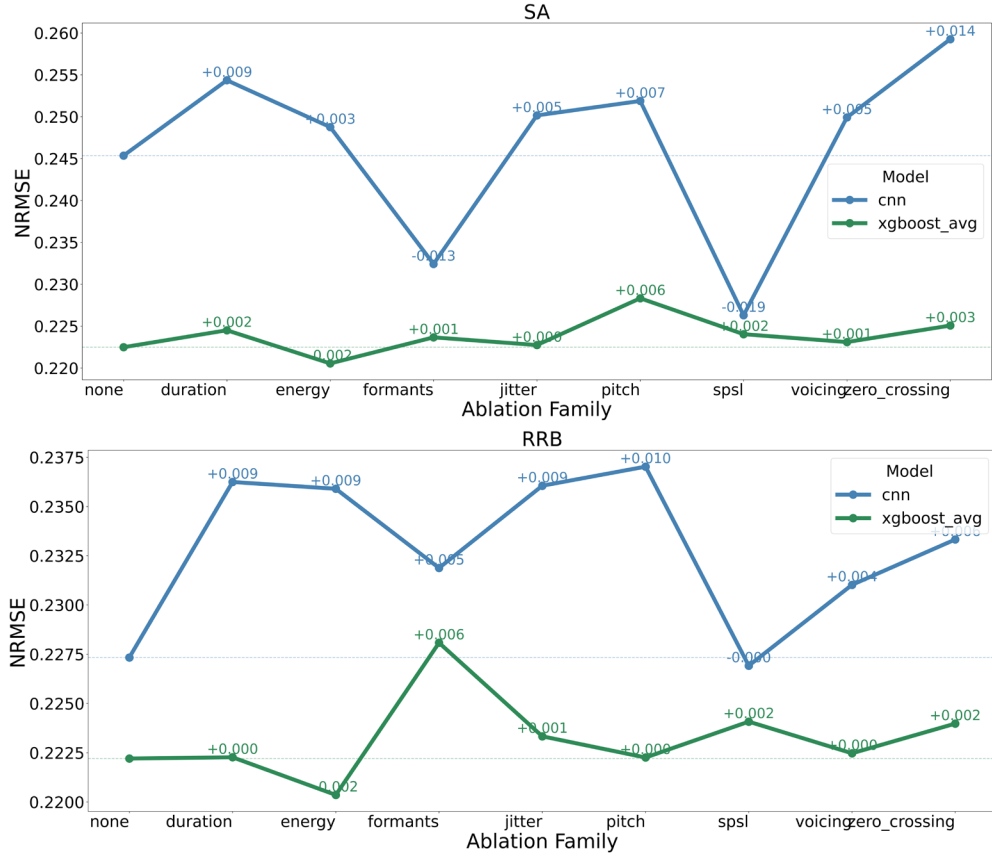


Figure 5. NRMSE for SA and RRB regression tasks under removal of feature groups.

To confirm that the CNN's performance is not limited by insufficient model capacity but rather by constraints imposed by the data representation, multiple higher-capacity architectures were evaluated, as shown in Figure 6. The medium variant doubles all filter and unit counts (128/256 convolutional, 256/128 dense), while the large variant further adds a third convolutional layer with an additional pooling stage and an expanded dense stack (512/256/128 units), resulting in approximately 4 $\times$ and 8 $\times$ more parameters than the baseline respectively.

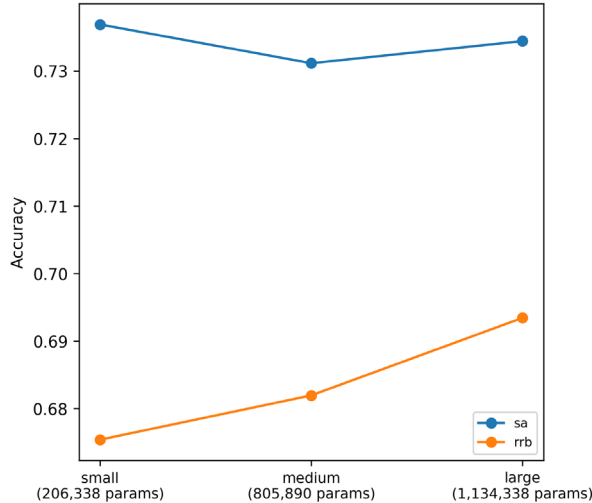


Figure 6. Effect of model complexity on accuracy on the SA task.

Despite increased model complexity, no performance gain was observed for SA, and only a modest ($\sim 2\%$) improvement was achieved for RRB. This suggests that the bottleneck lies in the data rather than model capacity.

Lastly, to assess the computational load of the evaluated models, Figure 7 presents the training and inference times. Training time refers to the total duration required for model fitting, while inference time represents the time needed to generate predictions on the full test set. All results are reported using the full feature set and are averaged over 20 runs for the SA case.

XGBoost is substantially cheaper computationally, requiring only around 1.4–1.5% of the CNN's training time for SA and RRB respectively, with a similarly small fraction of the CNN's inference time.

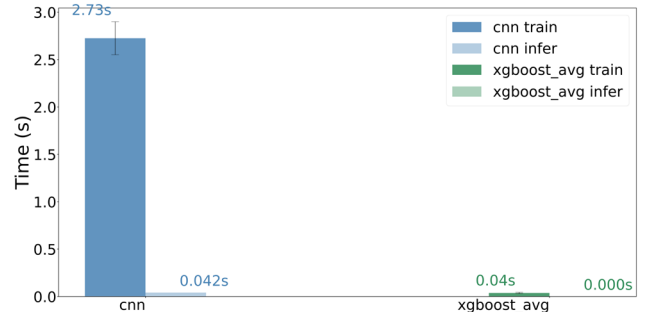


Figure 7. Training and inference time comparison between CNN and XGBoost models for the SA regression task.

V. CONCLUSION

The major findings of this work indicate that increased model complexity does not necessarily improve performance in speech-based ASD severity prediction. Simpler models such as XGBoost achieve comparable or superior results with substantially lower computational cost, and the results demonstrate that feature representation, particularly pitch and spectral characteristics, plays a more critical role than model architecture. Most speech feature families exhibit strong inter-correlations, with formant features providing relatively more independent information.

The speech correlations observed are broadly consistent with the existing literature. Elevated pitch as a marker of ASD severity is well-supported by meta-analytic evidence [6], [8], and by [5], which identifies pitch in voiced vocalizations as among the most important features for severity estimation. The co-occurrence of high F0 and steep spectral slope, interpreted here as reflecting breathy phonation, has received less direct attention elsewhere, and this interpretive framing represents a novel contribution of the present work. The negative correlation between jitter and severity is consistent with findings of greater vocal stability in ASD reported in [4]. For duration, the negative correlation observed here contrasts with meta-analytic evidence of longer vocalizations in ASD relative to typically developing individuals [6], though non-significant duration differences have also been reported [8]. For formant features, both [4] and [5] confirm their relevance to ASD classification and severity estimation. The paradox identified here, where atypical average formant positioning is combined with reduced within-speaker variance, is not discussed in prior work and represents a novel finding that connects vocal tract behavior to the broader behavioral definition of RRB.

This study has several limitations. The analysis relies on engineered and aggregated features rather than raw speech signals, which may limit the ability of more expressive models to capture temporal dynamics and long-range dependencies. Access to raw signals could enable models such as transformers to better exploit temporal structure, particularly over longer time horizons, though with significantly higher computational demand. Additionally, language differences may influence the findings; as noted in [7], the linguistic role of pitch in tonal languages may mask or alter ASD-related pitch differences in ways that do not apply to non-tonal language speakers. Individual physiological variation in vocal fold behavior and airflow generation also introduces variability in acoustic features, and subject-specific normalization may improve performance, particularly for features dependent on absolute values. A further limitation concerns the content-dependence of formant features (F11-F20) and zero-crossing rate (F33-F38): formant frequencies shift systematically with vowel identity, and ZCR varies with the proportion of fricatives and unvoiced consonants, meaning that observed differences may partly reflect phonemic content rather than intrinsic vocal characteristics. Although the structured task likely reduces this variability, phonemic content was not explicitly controlled. Finally, the overlap between validation and test sets, both drawn from the same child, introduces a dependency that should be noted, though this work is not

intended to demonstrate that speech features can predict ASD severity per se, a question already well-studied in the literature, but rather to investigate the development of more efficient models for that purpose.

Future work could explore dropping redundant feature families to build leaner, more efficient models that maintain predictive performance, as well as applying phonemic normalization strategies and extending the analysis to raw speech representations. The code is available at: <https://github.com/Mnaser95/ASD2>.

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